iv. The lower members are highly soluble in water due to the formation of intermolecular hydrogen bonding with water.

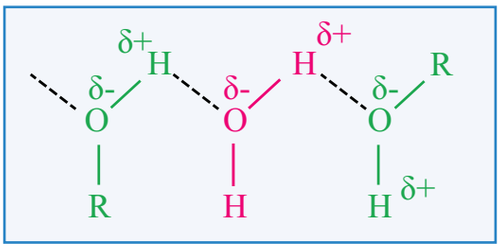


Table : Boiling point of alcohols in comparision with other organic compounds.

S.No.	Compound	Molecule formula	Molar mass	Boiling point (K)
1	Butane	CH_3-CH_2-CH_2-CH_3	58	272.5
2	Propanal	CH_3-CH_2-CHO	58	322
3	Methoxyethane	CH_3-O-CH_2-CH_3	60	283.8
4	Prapan – 1- ol	CH_3-CH_2-CH_2-OH	60	370.4
5	Propan – 2- ol	CH_3-CH(OH)-CH_3	60	355.5

Chemical properties of alcohols

Nucleophilic substitution reactions of alcohols

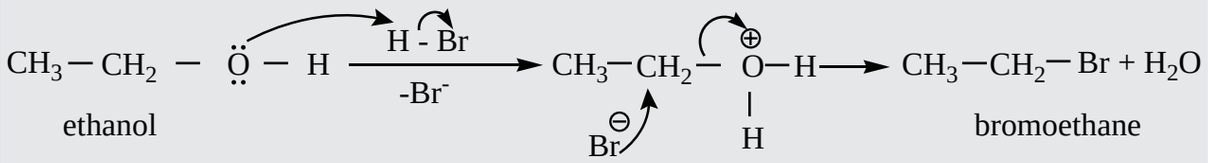
Alcohol has a strong basic leaving group (OH^-). So, –OH group is first converted into $-\overset{+}{O}H_2$ group by adding an acid. The $-\overset{+}{O}H_2$ group in the protonated alcohol can be easily displaced by a nucleophile such as Br^- to give alkyl halides.

Example: Alcohols undergo nucleophilic substitution reaction with hydro halic acids to form alkyl halides. In case of tertiary alcohols heating is required.

$$\underset{\text{Ethanol}}{CH_3-CH_2-OH} \xrightarrow[-\,H_2O]{HBr} \underset{\text{Bromoethane}}{CH_3-CH_2-Br}$$

Alkyl halide formation from primary alcohols follow S_N2 mechanism

Example



Nucleophilic attack of Br^- and leaving of H_2O takes place simultaneously.

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